

★ WEEK 2 · PART 2 · PRACTICE

Unit-Root Tests, Worked by Hand

Run the ADF, Then Confirm It With KPSS

The Dickey-Fuller statistic, its special cutoff, and the 2x2 grid

Part 2's decision tool, with the arithmetic shown. We run the ADF regression and compute its Dickey-Fuller statistic, compare it to the special critical value, then confirm with KPSS's opposite null - and use the two together to fix a series' order of integration.

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PART 2 · APPLICATIONS

Random Walk, or Mean Reverter?

Case A: run the ADF test on a spread - compute the Dickey-Fuller statistic and compare it to the special cutoff - to decide random walk versus stationary. Case B: confirm with the ADF-plus-KPSS 2x2 grid and pin down the order of integration. Then a Claude Code lab.

What Part 2 Told Us

Now we apply it - with numbers

1

A unit root is an AR coefficient of one

When ϕ equals one, the series is a random walk - non-stationary. The ADF test asks whether it does.

2

The ADF uses special critical values

Under a unit root, the statistic follows the Dickey-Fuller distribution, not the normal - so the cutoff is more negative.

3

ADF and KPSS confirm each other

They have opposite nulls, so running both and reading the 2x2 grid gives a confident, cross-checked verdict.

Everything here is a calculator exercise: one Dickey-Fuller statistic, one confirmatory grid.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a series and its ADF regression.
- Compute the statistic, or read the grid.
- Then decide stationarity.

What you should watch

- How the DF statistic beats a special cutoff.
- Why the cutoff is more negative than normal.
- How KPSS confirms the ADF.

The ADF Test

Random walk, or tradeable spread?

Before trading a spread, we must know it mean-reverts rather than wandering. The ADF regression tests exactly that: it estimates the coefficient on the lagged level and asks whether it's zero - a unit root. We compute the Dickey-Fuller statistic for a candidate spread, compare it to the special cutoff and reach a verdict

The setup

- A candidate pairs-trade spread.
- The ADF coefficient estimate is -0.18.
- Its standard error is 0.05.

What we'll do by hand

- Form the Dickey-Fuller statistic.
- Compare it to the special cutoff.
- Decide random walk or stationary.

The ADF Regression

Testing the lagged level

$$\Delta y_t = \alpha + \gamma y_{t-1} + \sum \delta_i \Delta y_{t-i} + \varepsilon_t; \quad H_0 : \gamma = 0 \text{ (unit root, since } \gamma = \phi - 1)$$

The setup

- Regress the change on the lagged level.
- The coefficient gamma is phi minus one.
- Extra lagged changes absorb short-run dynamics.

The null

- Gamma equals zero means phi equals one.
- That's a unit root - a random walk.
- Rejecting it means stationary.

The Dickey-Fuller Statistic

Coefficient over its standard error

$$\text{spread: } t = \frac{\hat{\gamma}}{\text{se}} = \frac{-0.18}{0.05} = -\mathbf{3.60} \quad (\hat{\phi} = 1 + \hat{\gamma} = 0.82, \text{ mean-reverting})$$

The calculation

- Minus 0.18 over 0.05 is minus 3.60.
- The implied phi is 0.82.
- Well below one - it mean-reverts.

What it suggests

- A strongly negative statistic.
- The lagged level pulls the change back.
- But we must use the right cutoff.

The Verdict

Against the special cutoff

$$-3.60 < \underbrace{-2.86}_{\text{DF 5\%}} < \underbrace{-1.65}_{\text{normal 5\%}} \Rightarrow \text{reject unit root} \Rightarrow \text{stationary (tradeable)}$$

Why special

- Under a unit root, the statistic isn't normal.
- It follows the Dickey-Fuller distribution.
- Shifted left, so the cutoff is more negative.

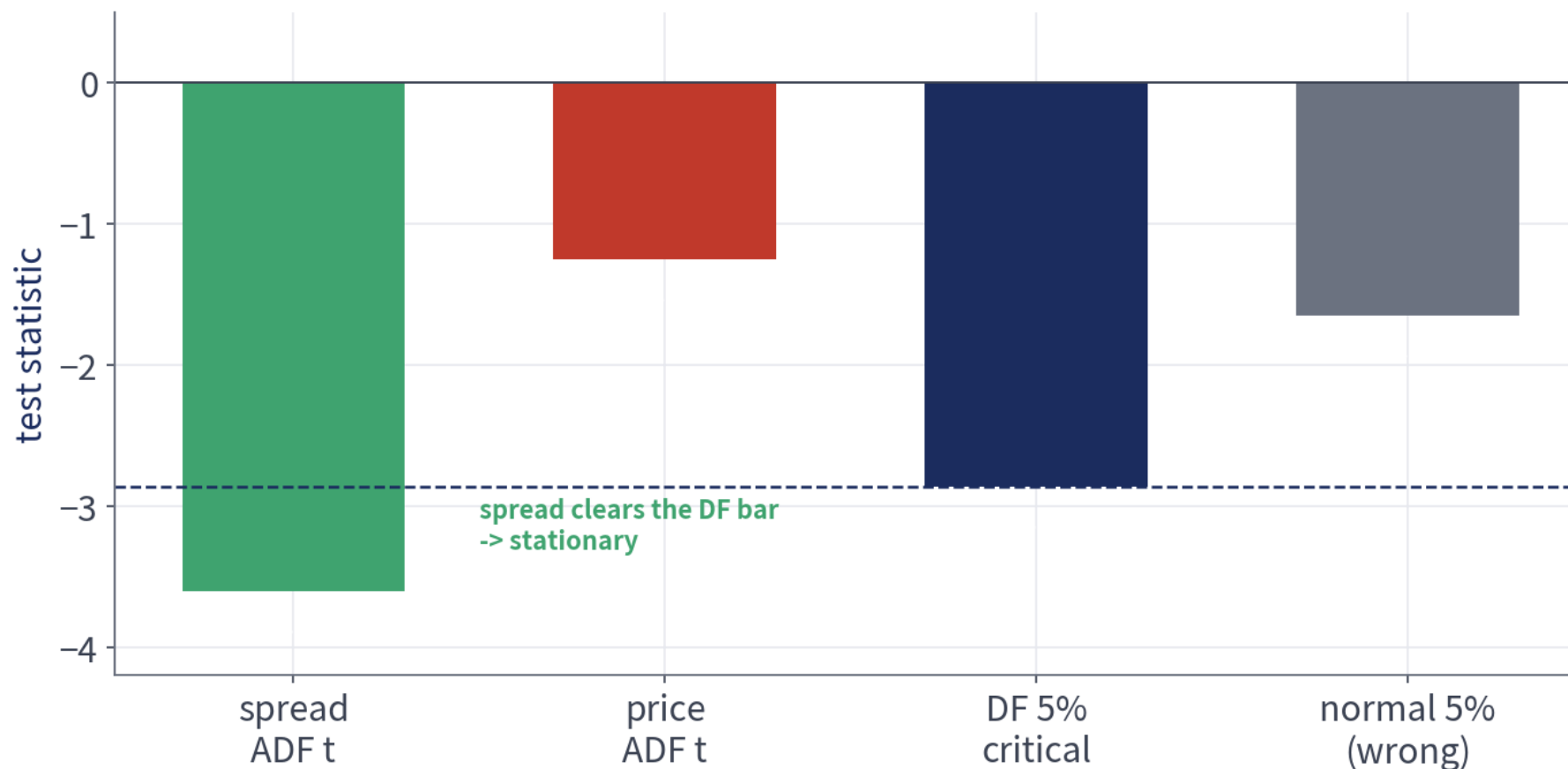
The decision

- At -3.60, it clears -2.86.
- Reject the unit root - the spread is stationary.
- The pair is tradeable.

Clearing the Right Bar

The spread's ADF statistic of -3.60 sits below the Dickey-Fuller cutoff of -2.86, so it's stationary. A price would give roughly -1.25, failing to clear it - a random walk. Note the DF cutoff is more negative than the normal -1.65: using the normal would wrongly declare random walks stationary

Case A: only the spread's ADF stat clears the Dickey-Fuller cutoff



Why the Diagnosis Matters

Spurious regression waits

The stakes

- Regressing two random walks looks significant.
- But the relationship is spurious.
- Non-stationarity fabricates false structure.

The safeguard

- Test for a unit root before regressing.
- Trade only spreads that reject it.
- The right cutoff prevents false confidence.

Confirm, and Count the Differences

ADF plus KPSS, and the order

One test can mislead, so we confirm. KPSS has the opposite null - it assumes stationarity - so pairing it with the ADF gives a cross-check. We read the 2x2 grid for a price and its returns, and in doing so pin down the order of integration: how many differences make the series stationary

The two tests

- ADF null: there is a unit root.
- KPSS null: the series is stationary.
- Opposite nulls, so they cross-check.

What we'll do by hand

- Read the grid for the price level.
- Then for its returns.
- Confirm the price is integrated of order one.

The Confirmatory Grid

Opposite nulls, read together

ADF H_0 : unit root $\perp$ KPSS H_0 : stationary (opposite nulls $\Rightarrow$ run both, read the grid)

Why two tests

- A single test can fail to reject weakly.
- The opposite null gives independent evidence.
- Agreement is a confident call.

Reading the corners

- ADF rejects and KPSS doesn't: stationary.
- ADF doesn't and KPSS rejects: unit root.
- Disagreement means inconclusive.

The Grid, Filled In

Price level versus returns

series	ADF	KPSS	verdict
price level	t=-1.25 (fail)	0.85 (reject)	unit root
returns	t=-8.50 (reject)	0.12 (fail)	stationary

For the price level, the ADF fails to reject and KPSS rejects stationarity - both point to unit root. For the returns, the ADF rejects and KPSS fails to reject - both point to stationarity. The two tests agree in each row, giving confident verdicts.

The Order of Integration

How many differences?

price level: ADF fails, KPSS rejects $\rightarrow$ difference $\rightarrow$ ADF rejects, KPSS fails $\Rightarrow$ price is $I(1)$

The logic

- The level is non-stationary - a unit root.
- One difference makes it stationary.
- So the price is integrated of order one.

In finance

- Prices are $I(1)$; returns are $I(0)$.
- That's why we model returns, not prices.
- The order tells you how much to difference.

The 2x2 Verdict

Where agreement is confident

Reading the grid

- Columns: does ADF reject?
- Rows: does KPSS reject?

ADF rejects (stationary) ADF fails (unit root)

Case B: opposite nulls -> the agreeing corners are confident calls

KPSS fails (stationary)	STATIONARY (confident)	inconclusive
KPSS rejects (unit root)	inconclusive	UNIT ROOT (confident)

price level -> UNIT ROOT cell; returns -> STATIONARY cell

confident;

- the off-diagonal is inconclusive.

Confirm Before You Difference

And difference the right amount

Why confirm

- A weak ADF result alone can mislead.
- KPSS's opposite null catches it.
- Agreement is what you act on.

Why the order matters

- Under-differencing leaves a unit root.
- Over-differencing injects noise.
- The order tells you exactly how much.

The Unit-Root Workflow

Test, confirm, difference

The routine is disciplined. Run the ADF to test for a unit root, comparing against its special Dickey-Fuller cutoff, not the normal. Confirm with KPSS, whose opposite null cross-checks the result - trust only the agreeing corners of the grid. Then difference just enough to reach stationarity, fixing the order of integration. That

The test (Case A)

- Spread ADF -3.60 clears -2.86.
- Stationary - tradeable.
- Price fails - a random walk.

The confirmation (Case B)

- Level: both tests say unit root.
- Returns: both say stationary.
- So the price is $I(1)$.

Reproduce the ADF and the Grid

Direct Claude Code to check your arithmetic

GOAL Run the ADF test, compare to the DF cutoff, and confirm with KPSS.

[INPUT] Prompt · Claude Code

```
Write and run part2_practice.py that:  
CASE A - ADF:  
  * simulate a stationary spread and a random-walk  
    price; run adfuller on each;  
  * print the statistic and compare to -2.86.  
CASE B - confirm and order:  
  * run adfuller and kpss on a price level and its  
    returns; print both statistics;  
  * report the 2x2 verdict and the order of integration.  
Then plot the ADF statistics and the 2x2 grid.
```

Check against the slides

- spread ADF $-3.60 < -2.86$.
- price ADF -1.25 (fail).
- level: ADF fail, KPSS reject.
- returns: ADF reject, KPSS fail $\rightarrow I(1)$.

Deliverable: part2_practice.py whose printed numbers match the ADF verdicts and the grid.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 What does the ADF null hypothesis say?

That the series has a unit root - it's non-stationary

Q2 Why use special critical values?

Under a unit root the statistic isn't normally distributed

Q3 Why pair ADF with KPSS?

Opposite nulls give a confident, cross-checked verdict

All three come from the ADF regression, its special cutoff, and the confirmatory grid.

Three Pitfalls to Avoid

Where unit-root testing trips people

Common mistakes

- "Compare the ADF statistic to normal cutoffs."
- "Failing to reject proves stationarity."
- "One test settles it."

The reality

- Use Dickey-Fuller critical values - not the normal.
- Failing to reject is weak evidence, not proof.
- Confirm with KPSS; trust the agreeing corners.

Two Calculations to Remember

Part 2 applications in one screen

1

The ADF statistic clears a special cutoff

The spread's Dickey-Fuller statistic was minus 0.18 over 0.05, or -3.60, implying a phi of 0.82. It cleared the special cutoff of -2.86 - more negative than the normal -1.65 - so we reject the unit root: the spread is stationary and tradeable.

2

ADF and KPSS confirm the order

For the price level, the ADF failed to reject and KPSS rejected stationarity - both say unit root. For the returns, the ADF rejected and KPSS failed to reject - both say stationary. The agreement pins the price as integrated of order one.

3

Test, confirm, then difference

Run the ADF against its special cutoff, confirm with KPSS's opposite null, and difference just enough to reach stationarity. That order of integration is the foundation every later model rests on.

Next — Part 3 practice: cointegration, when two non-stationary series share a stationary combination.